

EPISTEMIC WARRANT PROTOCOL

What an agent is justified in accepting — and why

EWP-0.2.0 release candidate · Policy reference-v2

MIT · Python 3.12+ · Canonical 14 · Pathological 12 · Hardening 25 · Goldens 26

EWP defines the deterministic boundary between what an AI agent's memory contains and what the agent is epistemically justified in accepting.

Memory is evidence, not truth.

Stores keep evidence. Warrant is computed. Action is a later gate.

Stores adapt to the protocol. The protocol does not inherit the store's epistemology.

1. What this is

EWP-0.2.0 is an epistemic protocol, not an action-authorization framework. Other projects use warrant to describe permission to act. EWP uses epistemic warrant to describe what an agent is justified in accepting. `may_act()` is deliberately a later gate.

```

Memory / Evidence Store
    |
    v
EvidenceView
    |
    v
warrant_now(view, policy, evaluated_at)
    |
    v
WarrantView
    |
    v
may_act(warrant, action, risk_policy)
    |
    v
MAY_ACT | REQUIRE_CONFIRMATION | DENY

```

Quick start

```

python3 tests/ci.py
python3 tests/report.py

```

```

from ewp.types import Policy
from ewp.warrant import warrant_now
from ewp.fixtures import fixture_verified_current, EVAL
print(warrant_now(fixture_verified_current(), Policy(), EVAL).warrant)

```

There is no packaged install yet. The repository itself is currently the reference implementation and conformance suite.

2. Why this exists

Agents keep conversations, observations, documents, tool results, inferred facts, summaries, and prior decisions. Remembering something is not the same as knowing it is true. Most systems store and retrieve those records. EWP asks: given the evidence available at time T, under policy P, what may this agent accept — and why?

Common collapse	EWP split
stored = true	stored != believed
retrieved = important	retrieved != complete
repeated = corroborated	repeated != independently sourced
summarized = verified	summarized != externally checked
current = warranted	store-local current != warrant
believed = safe to act on	warranted != authorized to act

The distinctions the protocol refuses to collapse.

3. The model

Evidence interchange

Stores expose assertions, evidence, provenance, lineage, conflicts, and verification records as an EvidenceView. Local current-fact, invalidation, ranking, and confidence machinery are inputs. They are not automatically agent beliefs.

Warrant evaluation

warrant_now(evidence_view, policy, evaluated_at) is deterministic. No I/O. No LLM. Same protocol version + same EvidenceView + same policy + same evaluated_at yields the same normative WarrantView. Values outside the closed enums are refused, not evaluated. Warrant is a function, not a stored truth field.

Decision gating

An agent may be justified in accepting that a customer requested cancellation and still need confirmation before cancelling a \$2M contract. Risk, reversibility, authorization, privacy, money, and safety belong in action policy.

4. WarrantView axes

EWP does not collapse state into TRUE, VERIFIED, or DISPUTED. A proposition can be externally checked and still disputed; accepted and stale; well evidenced with an open conflict.

Axis	Values
acceptance	UNACCEPTED / TENTATIVE / ACCEPTED
conflict	NONE / OPEN / RESOLVED
verification	NONE / INDIRECT / EXTERNAL / HUMAN
currency	CURRENT / STALE / SUPERSEDED
sufficiency	SUFFICIENT / INSUFFICIENT / DEGRADED

Verification is a record (method, source, scope, time, result, freshness policy), not a badge. result=opposes opens conflict and blocks ACCEPTED. result=inconclusive cannot raise EXTERNAL or HUMAN. Endogenous origin caps human and external methods at INDIRECT. Identity is declared subjects[], compared as exact ids: a check raises EXTERNAL or HUMAN only if neither side declares subjects or they share one. There is no family inference, and the scope string is never scraped. Currency is computed at evaluation time from checks that confer the chosen class. A later summary cannot refresh an old tool observation. History is not rewritten.

5. Lineage and the three confidences

Ten transformations of one source are not ten independent confirmations. The evaluator counts unique lineage_id values on assertions, evidence, and checks — not repetitions.

Kind	Meaning
Assertion confidence	How strongly the source or extractor stated the claim
Retrieval score	How relevant a record looks to this query
Warrant strength	How strongly policy permits acceptance

Implementations **MUST NOT** use retrieval relevance, repetition count, memory strength, or assertion confidence as substitutes for warrant.

6. Core invariants

- Assertions are not beliefs.
- Beliefs are not truth.
- Warrant is computed, not persisted as truth.
- Verification is evidence with method, scope, source, and time.
- Endogenous processing cannot manufacture external verification.
- Derivation cannot manufacture provenance.
- Contradiction must survive storage and retrieval.
- Incomplete retrieval must be visible as sufficiency=DEGRADED.
- Epistemic policy and action policy are separate.
- Semantically equivalent evidence must yield equivalent warrant independent of storage substrate.

Retrieval, summarization, dreaming, consolidation, reranking, repetition, graph propagation, LLM critique, and multi-agent agreement may reorganize. They may not turn verification=NONE into EXTERNAL without new external evidence. If search drops a contradictor, sufficiency is DEGRADED.

7. Platforms

OpenClaw

```
openclaw mcp add ewp --url http://127.0.0.1:8765/mcp
```

OpenClaw object	Role under EWP
Daily notes, transcripts	Raw lineage. Append. Do not rewrite.
USER.md	Working checkout of preferences. Regenerable.
MEMORY.md	Persona briefing from WarrantViews. Never authoritative.
Dreaming / promotion	Candidate generator. Must not raise verification class.
Local SQLite state	Cache. Not a second ledger.

Dreaming may rewrite MEMORY.md. EWP treats that rewrite as a new assertion, not as verification. Compression must not turn "X is disputed" into "X."

Claude, Codex, other MCP clients

Same contract. `python3 -m ewp.mcp_server` speaks the MCP stdio transport (newline-delimited JSON-RPC). `--http 127.0.0.1:8765` serves plain JSON-RPC POST `/mcp` for OpenClaw; it is not MCP Streamable HTTP. Every write requires the server-side ingest role (`--allow-ingest` on stdio, the token from `EWP_INGEST_TOKEN` on HTTP); without it the server is evaluate-only. Trusted origins also need `ingest_attestation=true`. Inline views are hypothetical: `ewp_warrant_now` demotes their trusted origins and `ewp_may_act` refuses them. `ewp_may_act` evaluates stored evidence at server time. The pre-freeze claim/confidence sketch stays in `historical/docs/MCP_CONTRACT.md` and is not the shipped façade.

```
python3 -m ewp.mcp_server --db ./ewp.sqlite
EWP_INGEST_TOKEN=... python3 -m ewp.mcp_server --http 127.0.0.1:8765 --db ./ewp.sqlite
openclaw mcp add ewp --url http://127.0.0.1:8765/mcp
```

Graphiti, Mem0, Particles, SQLite

Graphiti can be used as a temporal/entity evidence substrate or mirror. Inject `lineage_id` in episode metadata. `invalid_at` is store-local. `valid_at` is not a verification check. Search collapse marks the view DEGRADED. Every fixture must round-trip through the fake Graphiti and Mem0 paths with identical axes. Live Graphiti and Mem0 mappings live in `ewp/graphiti_client_adapter.py` and `ewp/mem0_adapter.py`; notes in `docs/implementer/LIVE_ADAPTERS.md`. Live graphiti-core 0.30.2 is not validated. Mem0 default origin is extract; retrieval score is not warrant. Particles is a good immutable substrate; writes go only through EWP's ingest role. SQLite and JSON prove store neutrality.

Graphiti adapts to the protocol. The protocol does not adapt to Graphiti.

8. Why EWP sits above the store

System	Primary concern	EWP adds
Graphiti / Zep	Temporal knowledge graph and retrieval	Store-independent warrant evaluation
Mem0	Agent memory storage and retrieval	Evidence lineage and deterministic warrant policy
OpenClaw native memory	Inspectable agent context and memory	Separation of persona, evidence, and warrant
EWP	Epistemic evaluation	Not a general-purpose memory store

9. Conformance

```
python3 tests/ci.py
python3 tests/report.py
python3 tests/runner.py
python3 tests/runner_pathological.py
```

CI enforces the fixture, evaluator, policy, golden, and implementer-pack lock hashes; all 26 goldens; all 51 fixtures through the codec, SQLite, JSON, Mem0, and fake Graphiti with identical axes and fields (Graphiti's listed losses aside); 12 invalid views refused; the hardening pack; the MCP façade; and an independent third evaluator. Changing a golden, the pack, the policy text, or the evaluator requires a protocol or policy version change, then `python3 tests/ci.py --write-lock`.

Class	Meaning
INGEST_LOSS	Store dropped assertions, sources, lineage, checks, or conflicts.
ADAPTER_MAP_LOSS	Store has the rows; EvidenceView is incomplete.
WARRANT_MISMATCH	Reconstructed view yields a WarrantView that is not the golden.
RETRIEVAL_LOSS	Search omitted evidence and did not mark DEGRADED.
EXPECTED_DIVERGENCE	Store current differs from warrant_now. Observation, not failure.

Pathological pack: false supersession, open contradiction, real temporal upgrade, same text / two lineages, three wordings / one lineage, store-current vs verified, weak invalidating strong, maintenance expiry that must not erase a check, retrieval-induced false consensus, extractor polarity flip on one lineage, canonical-without-verification, invalidation cycles.

10. Lock

```
Epistemic Warrant Protocol EWP-0.2.0
Policy: reference-v2
Canonical: 14/14   Pathological: 12/12   Hardening: 25/25   Invalid refused: 12/12
SQLite PASS      JSON PASS      Fake Graphiti PASS      Mem0 (fake client) PASS
graphiti-core 0.30.2 – NOT VALIDATED
fixture_set_sha256:
a6a9a1ea72608d70b695a5f7621978fca2c724dea877704c23ecea0ec8ed355
evaluator_set_sha256:
2fedf99023f4d17f85a44b8b8c89b825dcbcf4a5748c29f55aac5948b3dff025
policy_set_sha256:
90b1c819beffad833c5a43f4e68b014feea88d59bca5f60032684018da1c8896
golden_set_sha256:
1db7cab34639a87ce35c36635d4b6cffa46ba08855107fc9a04a4b89348a81c8
implementer_pack_sha256:
b13d6f36eb8e25419f45aab216991397fcff786468ee3e7c7596b76e72ee316c
```

A store that produces a different answer has an adapter or conformance problem, not a license to move the goldens.

11. What EWP is not

Not a vector database, knowledge graph, memory engine, truth oracle, LLM fact-checker, or authorization framework. It does not ask what is ultimately true. It asks a narrower, computable question: given this bounded evidence view, at this time, under this versioned policy, what may the agent accept — and why?

v0.2.0 tightened generic scope binding, refused future-dated checks at T, persisted SQLite view completeness, and narrowed the serialized WarrantView. Later 0.2.x hardening shipped the MCP façade with framed stdio and an ingest role, time-filtered assertions and evidence, and required subjects[] for scope caps. The 26 goldens stay. New stores may reveal adapter bugs; they do not redefine warrant.

Stores keep evidence. Warrant is computed. Action is a later gate.